



Selemun Abrha Teklu

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ABOUT ME

Recent First-Class Honours graduate in Computer Science and Engineering from Mekelle Institute of Technology, Mekelle University (CGPA: 3.87/4.00), with a strong research foundation in Machine Learning, Natural Language Processing, and efficient AI for low-resource languages. During a Machine Learning Researcher internship at Lesan AI (Berlin), I conducted comparative error analysis of production machine translation systems across healthcare and legal domains, strengthening my experience in literature review, model evaluation, and research methodology. I have practical expertise in Python, PyTorch, Hugging Face Transformers, and parameter-efficient fine-tuning techniques including LoRA and QLoRA, complemented by software engineering experience in production web development. I am seeking research-intensive graduate study to advance my expertise in efficient, low-resource AI systems and pursue long-term research

EDUCATION AND TRAINING

20/09/2019 – 26/06/2026 Mekelle, Ethiopia
BACHELOR DEGREE IN COMPUTER SCIENCE AND ENGINEERING MEKELLE INSTITUTE OF TECHNOLOGY

Some courses taken: Data Structure, Design and Analysis of Algorithms, Artificial Intelligence, Machine Learning, Natural Language Processing, Database Management Systems, Data communication, Software engineering, Mathematics for Natural Sciences, Applied Mathematics II, Applied Mathematics III, Discrete Mathematics, Probability and Statistics.

Website www.mu.edu.et | **Field of study** Computer Science and Engineering | **Final grade** 3.87/4.00 | **Type of credits** chrs | **Number of credits** 190 |

Thesis Parameter Efficient Fine-Tuning of NLLB for Bidirectional Tigrigna– English Healthcare Machine Translation: A Domain Adaptation Approach with Back Translation and RAG (Thesis)

WORK EXPERIENCE

MACHINE LEARNING RESEARCHER – LESAN AI (REMOTE) – 02/10/2025 – 31/01/2026 – BERLIN, GERMANY
Email: hello@lesan.ai | **Website:** <https://www.lesan.ai>

- Conducted comparative error analysis of Google, Microsoft, Meta NLLB, and Lesan for English–Tigrigna MT across healthcare and legal domains.
- Combined BLEU, chrF++, TER, COMET, and MQM-DQF human evaluation, identifying mistranslation (44.79%) and untranslated content (34.38%) as dominant errors.
- Identified critical medical and legal terminology errors, highlighting limitations of general-purpose MT and the need for domain-adapted models for low-resource languages.

WEB DEVELOPER MeMi Trading PLC

- 08/07/2025 – 07/11/2025, Mekelle , Ethiopia
- Coordinated the referral project project team and Contributed to the design, development, testing, and improvement of a Front end part of Referral Microservice integrated into Zemen bazaar a multivendor e-commerce platform (www.zemenbazaar.com).
 - Collaborated with cross-functional teams, coordinating tasks and facilitating communication between technical and business stakeholders to support effective decision-making.
 - Completed structured professional development training in entrepreneurship, psychometrics, leadership, and mindset development, strengthening teamwork and project execution skills.

TELECOM NETWORK & BASE STATION MAINTENANCE INTERN HUAWEI - ALKAN CIT(SUB-CON)

- 01/1/2025 – 20/04/2026 – MEKELLE, ETHIOPIA
- Performed hands-on inspection, operation, and maintenance of mobile base station components at Safaricom sites, ensuring proper site functionality and adherence to maintenance procedures.

- Collaborated with the Alkan Tigray maintenance team on end-to-end site maintenance workflows and real-world telecom infrastructure management.
- Observed and analyzed remote network monitoring, management, and configuration through the NOC, under the guidance of a Huawei engineer.

SKILLS

AI/ML

Python, NumPy, Pandas, Data Preprocessing, Feature Engineering | Scikit-learn, PyTorch, TensorFlow | Machine Learning, Deep Learning, NLP, Transformer Models, Model Training & Evaluation | Git/GitHub, VS Code, Docker, Jupyter Notebook, Google Colab

Web Development

React, Next.js, Tailwind Css, Rest Api Integration, postgresql, nextjs

LANGUAGE SKILLS

Mother tongue(s): **TIGRIGNA**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C2	C1	C1	C2
AMHARIC	C2	C2	C2	C2	C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

CERTIFICATIONS

Coursera, DeepLearning.AI, Udacity, stanford university, AWS

Online Certifications

Coursera certifications

- Machine Learning specialization-Coursera-([Link](#))
- Google Project Management specialization-Coursera-([Link](#))
- Deep Learning Specialization-Coursera-([Link](#))
- Natural Language processing specialization-Coursera-([Link](#))
- Data Analytics Professional Certificate-Coursera-([Link](#))
- CCNA : Introduction to Networks-CISCO Netacad website
- CCNA: Switching, Routing, and Wireless Essentials -CISCO Netacad website
- CCNA: Enterprise Networking, Security, and Automation -CISCO Netacad website
- Generative AI with LLMs-coursera-([Link](#))
- Pytorch for Deep Learning Specialization-coursera-([Link](#))

Mode of learning: Online

PROJECTS

10/02/2026 – CURRENT

Fine-Tuning Domain-Adapted Neural Machine Translation for Healthcare Domain: Bidirectional Tigrigna-English Translation Using NLLB (Thesis)

- Manually curated and validated a domain-specific parallel corpus of 1,837 clinical sentence pairs (615 English→Tigrigna, 1,222 Tigrigna→English) from Ayder Referral Hospital records and public medical guidelines, applying strict quality-filtering protocols.
- Fine-tuned NLLB-200-distilled-600M using LoRA, augmented with 3,000 back-translated synthetic pairs and Retrieval-Augmented Generation (RAG) for terminology-aware inference; deployed the fine-tuned model publicly on Hugging Face Spaces.
- Evaluated across BLEU, chrF++, TER, and COMET: English→Tigrigna BLEU improved from 1.64 to 31.75 (+1835%) with back-translation augmentation; Tigrigna→English BLEU improved from 10.33 to 18.19 (+76%), with chrF++ rising from 38.76 to 46.62. the fine-tuned model is deployed to hugging-face and see the live demo here([Link](#))

12/02/2025 – 20/06/2025

NLP based Geez Chatbot (mini-project)

- Developed a Geez-language conversational chatbot in a team by curating and annotating ~5,000 QA pairs and conducting research on low-resource NLP to guide model design.
- Fine-tuned and evaluated a multilingual mT5 transformer using efficient techniques (LoRA/PEFT) under limited resources, and documented the methodology and results for reproducibility.

Link <https://github.com/selemun-dev/geez-chatbot>

- Curated a 10,000+ sample multilingual speech dataset, building a Ge'ez-aware normalization pipeline, and implemented QLoRA adapters via Hugging Face's PEFT and Transformers libraries.
- Fine-tuned OpenAI's Whisper ASR model for Tigrinya using QLoRA, reducing WER from 1.133 to 0.375 while updating only a small fraction of model parameters.

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COMMUNITY ENGAGEMENTS

Volunteering Activities and Leadership

- Araya Zerihun Charity Club – General Secretary & Volunteer(10/10/2024-20/6/2026)
- Led and contributed to community service initiatives by coordinating aid distribution, organizing fundraising efforts,education and promoting volunteer engagement to support vulnerable groups.
- SNV – Volunteer, Mekelle Office (5/05/2024-6/04/2025)
- Supported digital data collection, analysis, monitoring, reporting, documentation, and community outreach.
 - Facilitated digital skills training and community sessions, promoting youth development, digital inclusion, and community resilience.
- Summer Coding Camp 2025 – Volunteer Teacher (08/07/2025 – 25/08/2025)
- Mentored and taught students aged between 14-18 in coding, mathematics, and gaming.
- Digital Signal Processing (DSP) Lab Tutor – MU-MIT(September 2025 – January 2026)
- Assisted and guided 3rd-year Computer Science and Engineering students during Digital Signal Processing (DSP) laboratory sessions, providing technical support in MATLAB-based practical exercises.
- Summer Tutorial Program Teacher-Tigray Development Association(01/07/2025 – 30/08/2025)
- Taught Mathematics and Introduction to Computer Science to Grade 8–12 students.

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SOFT SKILLS

soft skills

- Fast Learner and dedicated hard worker •Team Collaboration & Field Teamwork •Research-Oriented Problem Solving and Project Management •Adaptability in Field Environments •Technical Reporting & Communication •Communication and Leadership

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TRAININGS

07/07/2026 – 30/08/2026

INSA Cyber Talent Center Summer Camp - AI Engineering Track (5th Batch)

- Completed intensive AI engineering training covering Python for Data/Models, Mathematics for ML, NumPy/Pandas, EDA, and Supervised Learning, alongside practical exposure to big data and HPC systems.
- Contributed to a real-time cryptocurrency price-prediction pipeline using Kafka, HDFS, Spark, and ML models for end-to-end data streaming and prediction.
- Completed a cryptography practical assessment and sustained a demanding daily training schedule requiring discipline and consistency under pressure.